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An Expostion to the Theory of Varifolds and the Allard Regularity Theorem

End-Semester Report in Geometric Measure Theory (Fourth Semester)

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Abstract

This report presents an exposition of the theory of rectifiable varifolds and their role in geometric measure theory and the study of minimal surfaces. The primary objective is to develop enough of the theory to understand the statement and significance of the Allard Regularity Theorem, a fundamental regularity result for generalized minimal surfaces. We begin by introducing countably n-rectifiable sets as weak notions of surfaces and develop the concept of approximate tangent spaces, which allows many familiar objects from differential geometry to be extended to this nonsmooth setting. We then introduce rectifiable varifolds, which generalize rectifiable sets by incorporating multiplicity. After deriving an analogous first variation formula for rectifiable varifolds, we define stationary varifolds as weak analogues of minimal surfaces, and more generally, develop the notion of generalized mean curvature. Finally, we culminate with the Allard Regularity Theorem, which shows that sufficiently well-behaved varifolds at a given point are locally the graph of a C^1 function.

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0 Introduction

The aim of this report is to present an exposition of the theory of rectifiable varifolds, developing enough of the theory to ultimately state and understand the Allard Regularity Theorem, one of the central results in geometric measure theory and the theory of minimal surfaces. Broadly speaking, the theorem is a regularity result for generalized surfaces, and it is helpful to view it through an analogy with the theory of partial differential equations.

When studying general elliptic PDEs for instance, the usual strategy for proving existence of solutions follows a familiar paradigm: first establish the existence of a weak solution, often in a Sobolev space in the sense of distributions, and then appeal to a regularity theorem which upgrades this weak solution to a smooth one. In this sense, weak solutions provide a flexible framework in which existence can be established, while regularity theory explains when these weak objects are actually smooth.

The Allard Regularity Theorem plays a remarkably similar role for surfaces. Roughly speaking, a baby version of the theorem states that if a generalized surface (something that shall be made precise in the report) satisfies a suitable weak minimality condition, then it is in fact smooth at most points. Thus, although the surface is initially defined only in a weak measure-theoretic sense, the theorem shows that smooth geometric structure emerges automatically under suitable hypotheses.

To understand this result, we must first develop an appropriate weak notion of surface. Classical differential geometry works with smooth or at least C^1 submanifolds of $\mathbb{R}^{n+k}$, but such objects are far too restrictive for variational problems. Indeed, minimizing sequences for geometric functionals often fail to converge smoothly, and singularities may naturally appear in the limit. Consequently, we are led to enlarge the class of admissible surfaces.

The first step in this direction is the notion of a *countably n-rectifiable set*. These are subsets of $\mathbb{R}^{n+k}$ which can be covered, up to an $\mathcal{H}^n$-null set, by countably many images of Lipschitz maps

$$f_i : A_i \subset \mathbb{R}^n \to \mathbb{R}^{n+k}.$$

This provides a very weak notion of an n-dimensional surface while still retaining enough geometric structure to define meaningful tangent objects. One of the key ideas developed in this report is that many familiar notions from classical differential geometry continue to make sense in this setting, albeit in a weaker form. In particular, we shall define the notion of an *approximate tangent space*, which will play a central role throughout the report.

However, even countably rectifiable sets are not sufficiently general for many geometric variational problems. As we shall see through several examples, one would also like to encode multiplicity phenomena, where several “sheets” of a surface overlap. This leads naturally to the notion of a *rectifiable varifold*, which may be thought of heuristically as a rectifiable set equipped with a multiplicity function describing how many sheets are stacked over each point.

After developing the basic theory of rectifiable sets and varifolds in Section 1, we return to the main variational questions in Section 2. Recall that in the midsemester report we derived the classical first variation formula for C^2 submanifolds. Namely, if M is an n-dimensional C^2 submanifold and Φ_t is a compactly supported perturbation of M (see 2.5) with associated velocity vector field

$$X(x) = \left.\frac{\partial \Phi}{\partial t}\right|_{t=0}(x),$$

then

$$\left.\frac{d}{dt}\right|_{t=0} \mathcal{H}^n(\Phi_t(M)) = \int_M \operatorname{div}_M X \, d\mathcal{H}^n = -\int_M \langle X, H \rangle \, d\mathcal{H}^n,$$

where H denotes the mean curvature vector of M.

One of the main goals of this report is to extend this variational theory to rectifiable varifolds. At first sight, this may appear impossible: for generalized surfaces there is no obvious notion of tangent space, gradient, divergence, or curvature. The key observation is that the existence of approximate tangent spaces $\mathcal{H}^n$-almost everywhere allows us to define these objects in a weak sense. In particular, we shall derive an analogous first variation formula for rectifiable varifolds, which leads naturally to the notion of *generalized mean curvature*. This in turn allows us to define stationary varifolds, which serve as the appropriate weak analogue of minimal surfaces.

Finally, we shall study monotonicity formulas for stationary varifolds and varifolds with generalized mean curvature satisfying suitable bounds. These monotonicity formulas are among the most fundamental tools in geometric measure theory and ultimately lead to the statement of the Allard Regularity Theorem. Although we shall not prove the monotonicity theorems and Allard's theorem, the primary objective here is to develop the geometric and measure-theoretic framework necessary to properly understand its statement and significance.

The primary reference for this report is [3].

1 Rectifiable Sets and Varifolds

1.1 Countably n-Rectiable Sets

In the previous report, we worked primarily with C^2 submanifolds and derived the first variation formula in that setting. However, in geometric measure theory, one often needs to work with much weaker notions of surfaces, which may lack smooth structure altogether. This leads naturally to the class of *countably n-rectifiable sets*, which generalize smooth submanifolds while retaining enough structure to develop a meaningful calculus.

Definition 1.1. A set $M \subset \mathbb{R}^{n+k}$ is said to be *countably n-rectifiable* if there exist Lipschitz maps

$$f_i : \mathbb{R}^n \to \mathbb{R}^{n+k}, \qquad i = 1, 2, \ldots,$$

and a set $M_0 \subset \mathbb{R}^{n+k}$ with $\mathcal{H}^n(M_0) = 0$ such that

$$M \subset M_0 \cup \bigcup_{i=1}^{\infty} f_i(\mathbb{R}^n).$$

Writing $A_i := f_i^{-1}(M)$ and replacing M_0 by $M_0 \cap M$, we may express

$$M = M_0 \cup \bigcup_{i=1}^{\infty} f_i(A_i),$$

where $A_i \subset \mathbb{R}^n$ are arbitrary subsets (not necessarily measurable).

One can obtained an equivalent definition of the above notion which is extremely useful. To establish this, we first recall the following approximation result, and then state the equivalent characterization:

Lemma 1.2 (C^1 Approximation Theorem)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^{n+k}$ be a Lipschitz map. Then for every $\varepsilon > 0$, there exists a C^1 map $g : \mathbb{R}^n \to \mathbb{R}^{n+k}$ such that*

$$\mathcal{L}^n\Big(\{x : f(x) \neq g(x)\} \cup \{x : Df(x) \neq Dg(x)\}\Big) < \varepsilon.$$

Theorem 1.3. *A set $M \subset \mathbb{R}^{n+k}$ is countably n-rectifiable if and only if there exist C^1 n-dimensional submanifolds $\{N_i\}_{i=1}^{\infty}$ and a set N_0 with $\mathcal{H}^n(N_0) = 0$ such that*

$$M \subset N_0 \cup \bigcup_{i=1}^{\infty} N_i.$$

Proof. First suppose that such a decomposition exists. Each C^1 submanifold N_i can be covered by countably many coordinate charts $\varphi_{ij} : U_{ij} \subset \mathbb{R}^n \to N_i$, with U_{ij} bounded and φ_{ij} of class C^1, hence Lipschitz on U_{ij}. Therefore,

$$N_i = \bigcup_{j=1}^{\infty} \varphi_{ij}(U_{ij}),$$

and it follows that M is countably n-rectifiable.

Conversely, suppose M is countably n-rectifiable, so that

$$M \subset M_0 \cup \bigcup_{i=1}^{\infty} f_i(\mathbb{R}^n),$$

with $\mathcal{H}^n(M_0) = 0$ and each f_i Lipschitz.

By the C^1 approximation theorem, for each i we may choose a sequence of C^1 maps $\{g_{ij}\}_{j=1}^{\infty}$ such that $\mathcal{L}^n(\{x : f_i(x) \neq g_{ij}(x)\}) < 2^{-j}$. Let

$$B_{ij} := \{x \in \mathbb{R}^n : f_i(x) \neq g_{ij}(x)\}, \qquad B_i := \bigcap_{j=1}^{\infty} B_{ij}.$$

Then since f_i is Lipschitz, and g_{ij} is C^1, the set B_{ij} is Borel, hence $\mathcal{L}^n$ measurable. Thus, as B decreasing intersection of 2^{-j} measure sets, $\mathcal{L}^n(B_i) = 0$, and since f_i is Lipschitz, we also have $\mathcal{H}^n(f_i(B_i)) = 0$.

Thus

$$f_i(\mathbb{R}^n) \subset f_i(B_i) \cup \bigcup_{j=1}^{\infty} g_{ij}(\mathbb{R}^n),$$

and hence

$$M \subset M_0 \cup \bigcup_{i=1}^{\infty} f_i(B_i) \cup \bigcup_{i,j=1}^{\infty} g_{ij}(\mathbb{R}^n).$$

The union of the first two terms is still $\mathcal{H}^n$-null.

It remains to analyze the images of the C^1 maps g_{ij}. Note that $g_{ij}(\mathbb{R}^n)$ need not be C^1 submanifolds just because g_{ij} is C^1 as there could be points where the jacobian vanishes. However, this can be taken care of as shown below.

For each such map, define $C_{ij} := \{x \in \mathbb{R}^n : Jg_{ij}(x) = 0\}$, where Jg_{ij} denotes the Jacobian. By the area formula,

$$\mathcal{H}^n\Big(g_{ij}(C_{ij})\Big) = \int_{C_{ij}} Jg_{ij}(x)\, d\mathcal{H}^n(x) = 0.$$

On the set $\mathbb{R}^n \setminus C_{ij}$, the differential Dg_{ij} has full rank, so g_{ij} is an immersion. By the inverse function theorem, g_{ij} is locally a C^1 embedding, i.e. around each x in $\mathbb{R}^n \setminus C_{ij}$, there exists $U_x \subset \mathbb{R}^n \setminus C_{ij}$ open s.t $g_{ij}|_{U_x} : U_x \to g(U_x)$ is an embedding. i.e. Dg_{ij} has full rank on U_x and $g_{ij}|_{U_x}$ is a homeomorphism. Therefore, $g_{ij}(\mathbb{R}^n \setminus C_{ij})$ can be covered by countably many C^1 submanifolds.

Combining these observations, each $g_{ij}(\mathbb{R}^n)$ can be written as the union of a null set, namely the set where the Jacobian vanishes, and countably many C^1 submanifolds. Hence M admits the desired representation. $\square$

1.2 Approximate Tangent Spaces

In the previous section, we introduced countably n-rectifiable sets as a generalization of smooth submanifolds. Just as a C^1 submanifold admits a well-defined tangent space at each point, there is an analogous notion of *approximate tangent space* for much more general sets.

We will also see that the existence of approximate tangent spaces $\mathcal{H}^n$-almost everywhere characterizes countably n-rectifiable sets.

We begin by defining approximate tangent spaces in the setting of Radon measures. Before that, let us recall the notion of a Radon measure.

Definition 1.4 (Radon measure)**.** An outer measure μ on $\mathbb{R}^{n+k}$ is called a *Radon measure* if:

(i) μ is Borel regular, i.e. for every set $A \subset \mathbb{R}^{n+k}$ there exists a Borel set $B \supset A$ such that $\mu(A) = \mu(B)$,

(ii) μ is locally finite, i.e. $\mu(K) < \infty$ for every compact set $K \subset \mathbb{R}^{n+k}$.

Definition 1.5 (Approximate tangent space for measures)**.** Let μ be a Radon measure on $\mathbb{R}^{n+k}$. For $x \in \mathbb{R}^{n+k}$ and $\lambda > 0$, define the rescaled and translated measure $\mu_{x,\lambda}$ by

$$\mu_{x,\lambda}(A) := \lambda^{-n}\, \mu(x + \lambda A), \qquad A \subset \mathbb{R}^{n+k}.$$

We say that an n-dimensional subspace $P \subset \mathbb{R}^{n+k}$ is the *approximate tangent space* to μ at x, and write $T_x\mu = P$, if

$$\mu_{x,\lambda} \xrightarrow{v} \mathcal{H}^n \llcorner P \quad \text{as } \lambda \downarrow 0,$$

in the sense of *vague convergence of measures*, that is,

$$\int_{\mathbb{R}^{n+k}} f(y)\, d\mu_{x,\lambda}(y) \longrightarrow \int_P f(y)\, d\mathcal{H}^n(y) \quad \text{for all } f \in C_c(\mathbb{R}^{n+k}).$$

We now specialize this definition to sets. Let $M \subset \mathbb{R}^{n+k}$ be $\mathcal{H}^n$-measurable and suppose that

$$\mathcal{H}^n(M \cap K) < \infty \quad \text{for every compact set } K \subset \mathbb{R}^{n+k}.$$

Then $\mathcal{H}^n \llcorner M$ is a Radon measure, and we may define approximate tangent spaces for M via this measure.

Definition 1.6 (Approximate tangent space for sets)**.** Let $M \subset \mathbb{R}^{n+k}$ satisfy the above assumptions. We say that an n-dimensional subspace $P \subset \mathbb{R}^{n+k}$ is the *approximate tangent space* to M at a point $x \in \mathbb{R}^{n+k}$, and write $T_x M = P$, if

$$(\mathcal{H}^n \llcorner M)_{x,\lambda} \xrightarrow{v} \mathcal{H}^n \llcorner P \quad \text{as } \lambda \downarrow 0,$$

in the sense of vague convergence.
Equivalently,

$$\lim_{\lambda \to 0} \int_{(M-x)/\lambda} f(y)\, d\mathcal{H}^n(y) = \int_P f(y)\, d\mathcal{H}^n(y) \qquad \text{for all } f \in C_c(\mathbb{R}^{n+k}).$$

Remark 1.7 (Intuition)**.** The definition above can be understood as a *blow-up* procedure. The rescaled sets $(M - x)/\lambda$ represent zooming into the set M around the point x at finer and finer scales.

If M has a well-defined tangent behavior at x, then at small scales it should resemble a flat n-dimensional plane. The definition asserts precisely that, in the limit, the rescaled measures converge to $\mathcal{H}^n$ restricted to some subspace P. Thus, P captures the first-order geometric behavior of M near x.

It is also important to note that vague convergence was chosen over for instance weak convergence, because f being compactly supported is essential. We need to ensure f does not have any mass at infinity which might not be fully captured when zooming into x.
In the case of a C^1 submanifold, the approximate tangent space coincides with the usual tangent space, as shown later in this section.

Lemma 1.8. *If an approximate tangent space $T_x\mu$ exists at a point x, then it is unique.*

Proof. Suppose that P_1 and P_2 are two n-dimensional subspaces such that

$$\mu_{x,\lambda} \to \mathcal{H}^n \llcorner P_1 \quad \text{and} \quad \mu_{x,\lambda} \to \mathcal{H}^n \llcorner P_2$$

vaguely as $\lambda \downarrow 0$.
Then for every $f \in C_c(\mathbb{R}^{n+k})$,

$$\int f\, d(\mathcal{H}^n \llcorner P_1) = \lim_{\lambda \downarrow 0} \int f\, d\mu_{x,\lambda} = \int f\, d(\mathcal{H}^n \llcorner P_2).$$

Thus the measures $\mathcal{H}^n \llcorner P_1$ and $\mathcal{H}^n \llcorner P_2$ define the same linear functional on $C_c(\mathbb{R}^{n+k})$. By uniqueness in the Riesz representation theorem, it follows that $\mathcal{H}^n \llcorner P_1 = \mathcal{H}^n \llcorner P_2$, and hence $P_1 = P_2$. $\square$

Theorem 1.9. *Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold. Then for every $x \in M$, the approximate tangent space exists and*

$$T_xM \text{ (in the above sense)} = T_xM \text{ (in the classical sense)}.$$

Proof. Let $x \in M$. By translating and applying an orthogonal transformation, we may assume without loss of generality that $x = 0$ and that the classical tangent space at 0 is

$$T_0M = \mathbb{R}^n \times \{0\} \subset \mathbb{R}^{n+k}.$$

Let $f \in C_c(\mathbb{R}^{n+k})$. We aim to show that

$$\lim_{\lambda \downarrow 0} \int_{(M-0)/\lambda} f(y)\, d\mathcal{H}^n(y) = \int_{\mathbb{R}^n \times \{0\}} f(y)\, d\mathcal{H}^n(y).$$

Since M is a C^1 submanifold, there exists a chart $\varphi : B(0,1) \subset \mathbb{R}^n \to M$ such that $\varphi(0) = 0$, φ is C^1, and

$$D\varphi(0) = \begin{pmatrix} I_n \\ 0 \end{pmatrix}.$$

By possibly restricting and rescaling, we may assume φ is Lipschitz on $\overline{B(0,1)}$.

Assuming $\operatorname{supp}(f) \subset B(0,R)$, for λ small enough, repeated use of the area formula yields:

$$\begin{aligned}
\int_{M/\lambda} f(y)\, d\mathcal{H}^n(y) &= \frac{1}{\lambda^n} \int_{M \cap B(0,\lambda R)} f\left(\frac{y}{\lambda}\right)\, d\mathcal{H}^n(y) \\
&= \frac{1}{\lambda^n} \int f\left(\frac{\varphi(x)}{\lambda}\right) J\varphi(x)\, dx \\
&= \int f\left(\frac{\varphi(\lambda y)}{\lambda}\right) J\varphi(\lambda y)\, dy,
\end{aligned}$$

We now pass to the limit. Since f has compact support and φ is Lipschitz (hence $J\varphi$ is bounded), the integrand is uniformly bounded and supported in a fixed compact set (independent of λ for λ small). Hence we may apply the dominated convergence theorem when taking limit as λ tends to 0.
For each fixed $y \in \mathbb{R}^n$, we have

$$\lim_{\lambda \downarrow 0} \frac{\varphi(\lambda y)}{\lambda} = D\varphi(0)(y) = (y, 0),$$

and

$$\lim_{\lambda \to 0} J\varphi(\lambda y) = J\varphi(0) = 1.$$

Therefore,

$$\lim_{\lambda \downarrow 0} f\left(\frac{\varphi(\lambda y)}{\lambda}\right) J\varphi(\lambda y) = f(y, 0).$$

By dominated convergence,

$$\lim_{\lambda\to 0}\int_{\mathbb{R}^n} f\left(\frac{\varphi(\lambda y)}{\lambda}\right) J\varphi(\lambda y)\,dy = \int_{\mathbb{R}^n} f(y,0)\,dy.$$

Finally, applying area formula again, we obtain

$$\int_{\mathbb{R}^n} f(y,0)\,dy = \int_{\mathbb{R}^n\times\{0\}} f(z)\,d\mathcal{H}^n(z).$$

This completes the proof. $\square$

Finally, we state the fundamental existence result.

Theorem 1.10. *Let $M \subset \mathbb{R}^{n+k}$ be a countably n-rectifiable set s.t. $\mathcal{H}^n(M\cap K) < \infty$ for all compact sets $K \in \mathbb{R}^{n+k}$, i.e. $(\mathcal{H}^n \llcorner M)$ is a Radon measure. Then the approximate tangent space T_xM exists for $\mathcal{H}^n$-almost every $x \in M$.*

Before beginning the proof of this theorem, we must recall an important result seen earlier, namely the upper density theorem.

Definition 1.11 (Upper and lower n-densities)**.** Let μ be an outer measure on a metric space X, and let $A \subset X$. For $x \in X$, the n-dimensional upper and lower densities of μ relative to A at x are defined by

$$\Theta^{*n}(\mu, A, x) := \limsup_{r\downarrow 0} \frac{\mu\big(A\cap B_r(x)\big)}{\omega_n r^n},$$

and

$$\Theta^{n}_{*}(\mu, A, x) := \liminf_{r\downarrow 0} \frac{\mu\big(A\cap B_r(x)\big)}{\omega_n r^n},$$

respectively, where ω_n denotes the Lebesgue measure of the unit ball in $\mathbb{R}^n$.

Theorem 1.12 (Upper Density Theorem)**.** *Let μ be an outer measure on X such that all Borel sets are μ-measurable (for instance, $\mathcal{H}^n$), and let $A \subset X$ be μ-measurable with $\mu(A) < \infty$. Then*

$$\Theta^{*n}(\mu, A, x) = 0 \quad \textit{for } \mathcal{H}^n\textit{-almost every } x \in X\setminus A.$$

With this in mind, we can begin the proof for showing existence of approximate tangent space.

Proof of Theorem 1.10. By the C^1-rectifiability characterization in Theorem 1.3, there exist countably many n-dimensional C^1 submanifolds $\{N_j\}_{j=1}^{\infty}$ such that

$$M \subset N_0 \cup \bigcup_{j=1}^{\infty} N_j$$

and $\mathcal{H}^n(N_0) = 0$

Since we want to show existence of approximate tangent space $\mathcal{H}^n$ a.e., WLOG assume $N_0 = \emptyset$

Let $x \in M$. Then $\exists j$ such that $x \in N_j$. Note that since N_j is a C^1 submanifold, the classical tangent space $T_x N_j$ exists in the usual sense. We will show that $T_x M$ exists and equals $T_x N_j$ for $\mathcal{H}^n$-a.e. such x.

Let $f \in C_c(\mathbb{R}^{n+k})$. Consider the rescaled integrals

$$\int_{(M-x)/\lambda} f(y)\, d\mathcal{H}^n(y).$$

We decompose

$$M = (N_j \cup (M \setminus N_j)) \setminus (N_j \setminus M),$$

and hence

$$\int_{(M-x)/\lambda} f\, d\mathcal{H}^n = \int_{(N_j-x)/\lambda} f\, d\mathcal{H}^n + \int_{((M\setminus N_j)-x)/\lambda} f\, d\mathcal{H}^n. - \int_{((N_j\setminus M)-x)/\lambda} f\, d\mathcal{H}^n.$$

Since N_j is a C^1 submanifold and $x \in N_j$, by the previous theorem 1.9, we have

$$\lim_{\lambda \to 0} \int_{(N_j-x)/\lambda} f(y)\, d\mathcal{H}^n(y) = \int_{T_x N_j} f(y)\, d\mathcal{H}^n(y).$$

It remains to show that the error term vanishes. Write $E := M \setminus N_j$. Then

$$\int_{((E-x)/\lambda)} f(y)\, d\mathcal{H}^n(y) = \frac{1}{\lambda^n} \int_E f\Big(\frac{y-x}{\lambda}\Big)\, d\mathcal{H}^n(y).$$

Let $\operatorname{supp}(f) \subset B(0,R)$ and $\|f\|_\infty \le C$. Then

$$\left| \int_{((E-x)/\lambda)} f\, d\mathcal{H}^n \right| \le \frac{C}{\lambda^n} \mathcal{H}^n\Big(E \cap B(x, \lambda R)\Big).$$

Thus,

$$\left| \int_{((E-x)/\lambda)} f\, d\mathcal{H}^n \right| \le CR^n \cdot \frac{\mathcal{H}^n(E \cap B(x,\lambda R))}{\omega_n (\lambda R)^n} \cdot \omega_n,$$

where ω_n is the volume of the unit ball in $\mathbb{R}^n$.
The limsup of the term

$$\frac{\mathcal{H}^n(E \cap B(x,\lambda R))}{(\lambda R)^n}$$

as λ tends to 0 is (up to a constant) the upper n-density of $\mathcal{H}^n \llcorner E$ at x. Since $x \in M_j \subset N_j$, we have $x \notin E$. By the upper density theorem,

$$\Theta^{*n}(\mathcal{H}^n, E, x) = 0 \quad \text{for } \mathcal{H}^n\text{-a.e. } x \in M_j.$$

Therefore,

$$\lim_{\lambda \downarrow 0} \int_{((E-x)/\lambda)} f(y)\, d\mathcal{H}^n(y) = 0$$

for $\mathcal{H}^n$-a.e. $x \in M_j$.

Similarly, since $x \notin N_j \setminus M$, the third term in the integral expression also tends to 0 for $\mathcal{H}^n$ a.e.

Combining these observations, we obtain for $\mathcal{H}^n$-a.e. $x \in M_j$:

$$\lim_{\lambda \downarrow 0} \int_{(M-x)/\lambda} f(y)\, d\mathcal{H}^n(y) = \int_{T_x N_j} f(y)\, d\mathcal{H}^n(y).$$

Thus the approximate tangent space $T_x M$ exists and equals $T_x N_j$ for $\mathcal{H}^n$-a.e. $x \in M_j$. $\square$

An important observation from the above proof is made in the following corollary:

Corollary 1.13. *Let $M \subset \mathbb{R}^{n+k}$ be a countably n-rectifiable set such that $\mathcal{H}^n \llcorner M$ is a Radon measure. Suppose*

$$M \subset \bigcup_{j=1}^{\infty} N_j,$$

where each N_j is an n-dimensional C^1 submanifold. Then for $\mathcal{H}^n$-almost every $x \in M$, there exists an index j such that $x \in N_j$ and

$$T_x M = T_x N_j.$$

In particular, the approximate tangent space $T_x M$ is independent (up to $\mathcal{H}^n$-a.e. equality) of the choice of such a decomposition. That is, if also

$$M \subset \bigcup_{i=1}^{\infty} N_i',$$

with each N_i' an n-dimensional C^1 submanifold, then

$$T_x N_j = T_x N_i' = T_x M \quad \text{for } \mathcal{H}^n\text{-a.e. } x \in M,$$

whenever $x \in N_j \cap N_i'$.

Proof. The proof is an immediate consequence of the proof of the above Theoremn 1.9. $\square$

Finally, it is interesting to note that the converse of the Theorem 1.9 also holds, namely:

Theorem 1.14 (Converse characterization via approximate tangent spaces)**.** *Let $M \subset \mathbb{R}^{n+k}$ be an $\mathcal{H}^n$-measurable set such that*

$$\mathcal{H}^n(M \cap K) < \infty \quad \text{for every compact } K \subset \mathbb{R}^{n+k},$$

i.e. $\mathcal{H}^n \llcorner M$ *is a Radon measure. Suppose that the approximate tangent space* $T_x M$ *exists for* $\mathcal{H}^n$*-almost every* $x \in M$.
Then M *is countably* n*-rectifiable.*

Proof. We omit the proof, as it is rather involved and this result will not be needed in what follows. For a detailed argument, see [3]. □

We culminate this section by looking at a few examples for countably n-rectifiable sets, and more generally, Radon measures.

Example 1.15. Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold. Then M is countably n-rectifiable, and for every $x \in M$, the approximate tangent space exists and coincides with the classical tangent space:

$$T_x M \text{ (approximate)} = T_x M \text{ (classical)}.$$

This follows from the previous theorem.

Example 1.16. Let $I = [0, 1]$ and let $S = \partial(I^2) \subset \mathbb{R}^2$, the boundary of the unit square. Then S is clearly a countably 1-rectifiable set.

We examine the approximate tangent space at a corner, say $x = (0, 0)$. We claim that no 1-dimensional subspace P satisfies

$$(\mathcal{H}^1 \llcorner S)_{x,\lambda} \rightharpoonup \mathcal{H}^1 \llcorner P,$$

and hence the approximate tangent space (in the classical rectifiable sense) does not exist at $(0, 0)$.
However, the rescaled measures do admit a limit:

$$(\mathcal{H}^1 \llcorner S)_{(0,0),\lambda} \rightharpoonup \mathcal{H}^1 \llcorner P_1 + \mathcal{H}^1 \llcorner P_2,$$

where $P_1 = \{(t, 0) : t \in \mathbb{R}\}$ and $P_2 = \{(0, t) : t \in \mathbb{R}\}$ the x abd y axes respectively. Note that if we show the abover convergence holds, by uniqueness of limit in vague convergence, clearly S does not admit an approximate tangent space in the sense of rectifiable sets. We now justify the above claim.

To see this, parametrize the four edges using Lipschitz maps:

$$\varphi_1(t) = (t, 0), \quad \varphi_2(t) = (0, t), \quad \varphi_3(t) = (t, 1), \quad \varphi_4(t) = (1, t), \quad t \in [0, 1].$$

Then for $f \in C_c(\mathbb{R}^2)$,

$$\int f(y)\, d(\mathcal{H}^1 \llcorner S)_{(0,0),\lambda}(y) = \frac{1}{\lambda} \int_S f\left(\frac{z}{\lambda}\right)\, d\mathcal{H}^1(z).$$

We split this into four integrals corresponding to the four edges by the area formula. For the bottom edge, we get:

$$\frac{1}{\lambda}\int_0^1 f\left(\frac{(t,0)}{\lambda}\right)\,dt = \int_0^{1/\lambda} f(s,0)\,ds \ \to\ \int_{\mathbb{R}} f(s,0)\,ds,$$

as $\lambda \to 0$ by DCT which can be applied as f is compactly supported. Similarly, for the left edge,

$$\frac{1}{\lambda}\int_0^1 f\left(\frac{(0,t)}{\lambda}\right)\,dt \to \int_{\mathbb{R}} f(0,s)\,ds.$$

The contributions from the top and right edges vanish in the limit, as can easily be checked by the same calculation.

Thus the limit measure is

$$\mathcal{H}^1 \llcorner P_1 + \mathcal{H}^1 \llcorner P_2,$$

which is not supported on a single line. Hence the approximate tangent space at $(0,0)$ does not exist, however it does exist in the sense of Radon measures.

We end this section by motivating the definition of approximate tangent spaces with multiplicity, and prove similar existence results in the next section after we have defined varifolds.

Remark 1.17 (A limitation of the definition)**.** In the previous definition, the approximate tangent space is required to be a single n-dimensional subspace. This turns out to be too restrictive in situations where multiple “sheets” of a set overlap.

To illustrate this, consider the following example.

Example 1.18. Let $M = C_1 \cup C_2 \subset \mathbb{R}^2$, where C_1 and C_2 are two C^1 curves which are tangent to each other at the point $x = (0,0)$, and have the same tangent line at that point (for instance, two circles tangent at $(0,0)$ with horizontal tangent).

Then for each $i = 1,2$, the approximate tangent space of C_i at x exists and is equal to the same line P (the common tangent line).

Now consider $M = C_1 \cup C_2$. For $f \in C_c(\mathbb{R}^2)$, we have

$$\int_{(M-x)/\lambda} f(y)\,d\mathcal{H}^1(y) = \int_{(C_1-x)/\lambda} f(y)\,d\mathcal{H}^1(y) + \int_{(C_2-x)/\lambda} f(y)\,d\mathcal{H}^1(y).$$

Passing to the limit as $\lambda \downarrow 0$, each term converges to

$$\int_P f(y)\,d\mathcal{H}^1(y),$$

and hence

$$\int_{(M-x)/\lambda} f(y)\,d\mathcal{H}^1(y) \longrightarrow 2\int_P f(y)\,d\mathcal{H}^1(y).$$

Thus, the blow-up limit is not $\mathcal{H}^1 \llcorner P$, but rather $2\,\mathcal{H}^1 \llcorner P$.

Therefore, in the sense of the previous definition, the approximate tangent space does not exist at x. However, geometrically, it is natural to interpret this as two sheets overlapping along the same tangent line.

The above example shows that the notion of approximate tangent space should allow for a *multiplicity* factor, which records how many sheets of the set are present at a point. This leads to the following refined definition.

Definition 1.19 (Approximate tangent space with multiplicity)**.** Let μ be a Radon measure on $\mathbb{R}^{n+k}$. We say that an n-dimensional subspace $P \subset \mathbb{R}^{n+k}$ is the *approximate tangent space* to μ at x with multiplicity $\theta(x) > 0$ if

$$\mu_{x,\lambda} \longrightarrow \theta(x)\,\mathcal{H}^n \llcorner P \quad \text{as } \lambda \downarrow 0,$$

in the sense of vague convergence, that is,

$$\int f\, d\mu_{x,\lambda} \longrightarrow \theta(x) \int_P f\, d\mathcal{H}^n \qquad \text{for all } f \in C_c(\mathbb{R}^{n+k}).$$

Remark 1.20. In the example above, the approximate tangent space at $(0,0)$ exists in this generalized sense, with tangent line P and multiplicity $\theta(0,0) = 2$.

1.3 General and Rectifiable Varifolds

In the previous section, we saw that the definition of the approximate tangent space developed prior was too restrictive, and motivated the need to consider approximate tangent spaces with multiplicity.

We begin this section with an example which shall illustrate why the notion of countably n recitifiable sets is also restrictive, and motivate why one may want to consider a multiplicity function as well for the rectifiable set M, leading to the defintion of a varifold in this section. We shall then revisit this idea of tangent space with multiplicity in the context of rectifiable varifolds, and prove similar existence results as we did for countably n-rectifiable sets

Example 1.21 (Multiplicity arising from overlapping sheets)**.** Consider, for each $k \in \mathbb{N}$, the set $M_k \subset \mathbb{R}^2$ given by the union of two lines

$$M_k = L_k^+ \cup L_k^-,$$

where

$$L_k^\pm = \{(x, \pm\tfrac{1}{k}x) : x \in \mathbb{R}\}.$$

Each M_k is a union of two distinct lines intersecting at the origin.

As $k \to \infty$, the slopes $\pm\frac{1}{k}$ tend to 0, and the two lines become increasingly close to the x-axis. In a heuristic sense, the sets M_k approach the single line

$$M = \{(x, 0) : x \in \mathbb{R}\},$$

but with two sheets overlapping.

Intuitively, one would feel that defining the limiting set simply as the x-axis would not truly capture the geometric situation. Indeed, for if we simply had L_k^+, i.e. one line, the limiting configuration would be the same which feels unsatisfactory. However, it is unclear how one can describe two sheets of the x-axis using only the language of countably n-rectifiable sets.

From the point of view of varifolds, this limiting configuration is naturally described by the rectifiable varifold $v(M, \theta)$ with multiplicity $\theta \equiv 2$, reflecting the presence of two coinciding sheets. This last paragraph will make more sense once we have defined and explored varifolds, after which we shall return to this example, also making the notion of this heuristic "convergence" more precise.

We begin first by defining general varifolds. Although we shall work with a more restrictive type of varifolds, namely rectifiable ones, we regardless define this more general notion.

Definition 1.22 (Grassmannian)**.** Let $G(n+k, n)$ denote the collection of all n-dimensional linear subspaces of $\mathbb{R}^{n+k}$.
For $S, T \in G(n+k, n)$, define

$$d(S, T) = \left(\sum_{i,j=1}^{n+k} \left(p_{ij}^S - p_{ij}^T \right)^2 \right)^{1/2},$$

where p^S, p^T denote the orthogonal projections onto S and T respectively, and

$$p_{ij}^S = \langle e_i, p^S(e_j) \rangle.$$

where e_i denotes the standard orthonormal basis in $\mathbb{R}^{n+k}$

This metric induces a natural topology on $G(n+k, n)$.
For any set $A \subset \mathbb{R}^{n+k}$, we define

$$G_n(A) := A \times G(n+k, n).$$

We are now in a position to define varifolds.

Definition 1.23 (Varifold and Weight Measure)**.** Let $U \subset \mathbb{R}^{n+k}$ be open. An *n-varifold* in U is a Radon measure V on $G_n(U) = U \times G(n+k, n)$.

Thus, a varifold assigns mass not only to points in space, but also to choices of n-dimensional tangent planes.

Associated to every varifold is a measure on the ambient space, namely the weight measure. The *weight measure* μ_V on U is defined by

$$\mu_V(A) = V\Big(\pi^{-1}(A)\Big), \qquad A \subset U,$$

where $\pi : G_n(U) \to U$ is the projection $\pi(x, S) = x$.
The total mass of V is then defined by

$$\mathbf{M}(V) := \mu_V(U) = V(G_n(U)).$$

If $A \subset U$ is Borel, we denote by $V \llcorner G_n(A)$ the restriction of V to $G_n(A)$, i.e.

$$(V \llcorner G_n(A))(B) = V\Big(B \cap G_n(A)\Big), \qquad B \subset G_n(U).$$

In this report, we shall only be interested in a special class of varifolds, namely rectifiable varifolds which are those arising from rectifiable sets. This class by itself is general enough and produces abundant examples.

Definition 1.24 (Rectifiable varifold)**.** Let $U \subset \mathbb{R}^{n+k}$ be open. Let $M \subset U$ be a countably n-rectifiable set which is $\mathcal{H}^n$-measurable, and let

$$\theta : M \to (0, \infty)$$

be an $\mathcal{H}^n$-measurable function such that

$$\int_{M \cap K} \theta \, d\mathcal{H}^n < \infty \quad \text{for every compact } K \subset U.$$

i.e. $\theta \, d\mathcal{H}^n|_M$ is a Radon measure. Then the pair (M, θ) determines a *rectifiable n-varifold*, denoted $V = v(M, \theta)$, where θ is called the *multiplicity function*.

Two pairs (M, θ) and $(\widetilde{M}, \widetilde{\theta})$ define the same rectifiable varifold if they agree up to $\mathcal{H}^n$-null sets and $\theta = \widetilde{\theta}$ almost everywhere on their intersection.

We now describe how such a rectifiable varifold may be viewed as a general varifold.

Definition 1.25 (Associated measure on $G_n(U)$)**.** Let $V = v(M, \theta)$ be a rectifiable varifold. Define a measure V on $G_n(U)$ by

$$V(A) = \int_M \mathbf{1}_A(x, T_x M) \, \theta(x) \, d\mathcal{H}^n(x), \qquad A \subset G_n(U),$$

where $T_x M$ denotes the approximate tangent space to M at x, defined for $\mathcal{H}^n$-almost every $x \in M$.

In other words, the varifold V is obtained by assigning to each point $x \in M$ the tangent plane T_xM together with weight $\theta(x)$.

It follows immediately from the definition that the associated weight measure satisfies

$$\mu_V = \theta\, \mathcal{H}^n \llcorner M.$$

Henceforth, we shall only be discussing rectifiable varifolds.

We now extend the notion of approximate tangent space to rectifiable varifolds.

Definition 1.26 (Approximate tangent space of a varifold)**.** Let $U \subset \mathbb{R}^{n+k}$ be open, and let

$$V = v(M, \theta)$$

be a rectifiable n-varifold in U. Let $x \in M$.
We say that an n-dimensional subspace $P \subset \mathbb{R}^{n+k}$ is the *approximate tangent space of* V *at* x if for every $f \in C_c(\mathbb{R}^{n+k})$,

$$\lim_{\lambda \to 0} \int_{\eta_{x,\lambda}(M)} f(y)\, \theta(x + \lambda y)\, d\mathcal{H}^n(y) = \theta(x) \int_P f(y)\, d\mathcal{H}^n(y),$$

where $\eta_{x,\lambda}(y) = \frac{y-x}{\lambda}$ denotes the rescaling map.
In other words, equivalently it follows that

$$(\mu_V \llcorner M)_{x,\lambda} \xrightharpoonup{v} \theta(x)\mathcal{H}^n \llcorner P$$

where $\mu_V = \theta\, \mathcal{H}^n|_M$

If such a subspace exists, it is uniquely determined and we denote it by T_xV (or T_xM when no ambiguity arises).

Remark 1.27. In the case $\theta \equiv 1$, this definition reduces to the usual notion of approximate tangent space for rectifiable sets. In particular, the two notions agree $\mathcal{H}^n$-almost everywhere.

As in the case of rectifiable sets, one can show that approximate tangent spaces exist $\mathcal{H}^n$-almost everywhere for rectifiable varifolds. We will return to this shortly. Before that, we give an intuitive picture and an example to elucidate the role of the multiplicity function θ.

Remark 1.28 (Role of the multiplicity function)**.** The multiplicity function θ encodes how much "mass" of the varifold is concentrated at a given point.

In the special case $\theta \equiv 1$, the rectifiable varifold $v(M, 1)$ may be identified directly with the set M itself: at $\mathcal{H}^n$-almost every point $x \in M$, the varifold simply records the approximate tangent space T_xM with unit weight.

In general, the function θ may be interpreted as describing the number of overlapping sheets of the surface. More precisely, if $\theta(x) = k$ and the approximate tangent space T_xM exists at x, then one may think of the varifold as consisting of k copies of the plane T_xM stacked on top of each other at the point x.

Thus, while the underlying set M captures the geometric support of the varifold, the multiplicity function θ records how many times the surface is counted at each point.

Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$. We now record a few useful definitions:

Definition 1.29 (Weight measure)**.** The *weight measure* associated to V is given by

$$\mu_V = \theta\, \mathcal{H}^n \llcorner M,$$

where we adopt the convention that $\theta \equiv 0$ on $U \setminus M$.

Thus, for any $\mathcal{H}^n$-measurable set $A \subset U$,

$$\mu_V(A) = \int_{A \cap M} \theta \, d\mathcal{H}^n.$$

Definition 1.30 (Mass)**.** The *mass* of the varifold V is defined by

$$\mathbf{M}(V) = \mu_V(U) = \int_M \theta \, d\mathcal{H}^n.$$

Definition 1.31 (Support)**.** The *support* of V is defined as

$$\operatorname{spt} V := \operatorname{spt} \mu_V,$$

that is, the set of all points $x \in U$ such that $\mu_V(B_r(x)) > 0$ for all $r > 0$.

Definition 1.32 (Restriction)**.** If $A \subset U$ is $\mathcal{H}^n$-measurable, we define the restricted varifold

$$V \llcorner A := v(M \cap A, \theta|_A).$$

Definition 1.33 (Weak convergence)**.** Let $V_k = v(M_k, \theta_k)$ be a sequence of rectifiable varifolds in U. We say that $V_k \to V$ if the associated weight measures converge weakly as Radon measures, i.e.

$$\mu_{V_k} \rightharpoonup \mu_V \quad \text{in } U.$$

Remark 1.34. Let us now return to the example considered in the beginning of this section, namely 1.21. We can think of $M_k = L_k^+ \cup L_k^-$, where

$$L_k^\pm = \{(x, \pm\tfrac{1}{k}x) : x \in \mathbb{R}\},$$

as a varifold $V_k = v(M_k, \theta_k)$ with multiplicity function $\theta_k \equiv 1$. We claim that V_k converges (in the sense of weight measures) to the varifold $V = v(M, \theta)$, where

$$M = \{(x, 0) : x \in \mathbb{R}\}, \qquad \theta \equiv 2.$$

Note that $\mu_{V_k} = \mathcal{H}^1 \llcorner M_k$ and $\mu_V = 2\,\mathcal{H}^1 \llcorner M$. We show that $\mu_{V_k} \rightharpoonup \mu_V$ (vaguely) as Radon measures.

Let $f \in C_c(\mathbb{R}^2)$. Then

$$\int f\, d\mu_{V_k} = \int_{L_k^+} f\, d\mathcal{H}^1 + \int_{L_k^-} f\, d\mathcal{H}^1.$$

Parametrize the two lines by

$$\gamma_k^{\pm}(x) = (x, \pm\tfrac{1}{k}x), \qquad x \in \mathbb{R}.$$

Hence, by the area formula,

$$\int_{L_k^{\pm}} f\, d\mathcal{H}^1 = \int_{\mathbb{R}} f\Big(x, \pm\tfrac{1}{k}x\Big)\,\sqrt{1 + \tfrac{1}{k^2}}\, dx.$$

Therefore,

$$\int f\, d\mu_{V_k} = \sqrt{1 + \tfrac{1}{k^2}}\,\int_{\mathbb{R}} \Big[f\Big(x, \tfrac{1}{k}x\Big) + f\Big(x, -\tfrac{1}{k}x\Big)\Big] dx.$$

Since f has compact support, the integrals are effectively over a fixed bounded set. Moreover,

$$f\Big(x, \pm\tfrac{1}{k}x\Big) \to f(x, 0) \quad \text{pointwise as } k \to \infty,$$

and $\sqrt{1 + \frac{1}{k^2}} \to 1$. The integrand is uniformly bounded by $2\|f\|_\infty\sqrt{2}$, so by the dominated convergence theorem,

$$\int f\, d\mu_{V_k} \longrightarrow \int_{\mathbb{R}} \Big[f(x, 0) + f(x, 0)\Big] dx = 2\int_{\mathbb{R}} f(x, 0)\, dx.$$

Finally, by the area formula,

$$2\int_{\mathbb{R}} f(x, 0)\, dx = \int_M f\,(2\, d\mathcal{H}^1) = \int f\, d\mu_V.$$

Thus $\mu_{V_k} \rightharpoonup \mu_V$, as claimed.

We now return to the question of existence of approximate tangent spaces, this time for varifolds. Recall that in 1.10, we showed this for countably n-rectifiable sets, i.e. varifolds with multiplicity function $\theta \equiv 1$. We now extend this to varifolds:

Theorem 1.35. *Let $U \subset \mathbb{R}^{n+k}$ be open, and let $V = v(M, \theta)$ be a rectifiable n-varifold in U. Then the approximate tangent space $T_x V$ exists for $\mathcal{H}^n$-almost every $x \in M$.*

Proof. By Theorem 1.10, the approximate tangent space $T_x M$ exists for $\mathcal{H}^n$-almost every $x \in M$. Moreover, since θ is locally integrable with respect to $\mathcal{H}^n \llcorner M$, the Lebesgue differentiation theorem implies that for $\mathcal{H}^n$-almost every $x \in M$,

$$\lim_{\lambda \downarrow 0} \frac{1}{\lambda^n} \int_{M \cap B_\lambda(x)} |\theta(y) - \theta(x)| \, d\mathcal{H}^n(y) = 0.$$

Fix such a point $x \in M$, i.e. where the above identity holds by Lebesgue differentiation and $P = T_x M$ exists. We show that P is also the approximate tangent space of the varifold V at x with multiplicity $\theta(x)$.

Recall that $n_{x,\lambda}(A) := (A - x)/\lambda$ and let $f \in C_c(\mathbb{R}^{n+k})$. We now estimate:

$$\left| \int_{\eta_{x,\lambda}(M)} f(z)\theta(x + \lambda z) \, d\mathcal{H}^n(z) - \theta(x) \int_P f \, d\mathcal{H}^n \right|.$$

Using the triangle inequality, this is bounded above by

$$\int_{\eta_{x,\lambda}(M)} f(z)\Big(\theta(x + \lambda z) - \theta(x)\Big) \, d\mathcal{H}^n(z) + \theta(x) \left| \int_{\eta_{x,\lambda}(M)} f(z) \, d\mathcal{H}^n(z) - \int_P f \, d\mathcal{H}^n \right|.$$

Since x is a point where $T_x M$ exists, the second term converges to 0 as $\lambda \downarrow 0$.

For the first term, let $K = \operatorname{spt}(f)$, which is compact. Then

$$\left| \int_{\eta_{x,\lambda}(M)} f(z)\Big(\theta(x + \lambda z) - \theta(x)\Big) \, d\mathcal{H}^n(z) \right|$$

$$\leq \|f\|_\infty \int_{\eta_{x,\lambda}(M) \cap K} |\theta(x + \lambda z) - \theta(x)| \, d\mathcal{H}^n(z).$$

Changing variables back to $y = x + \lambda z$,

$$= \frac{\|f\|_\infty}{\lambda^n} \int_{M \cap (x + \lambda K)} |\theta(y) - \theta(x)| \, d\mathcal{H}^n(y).$$

Since K is compact, there exists $R > 0$ such that $K \subset B_R(0)$. Hence

$$x + \lambda K \subset B_{R\lambda}(x),$$

and therefore

$$\leq \frac{\|f\|_\infty}{\lambda^n} \int_{M \cap B_{R\lambda}(x)} |\theta(y) - \theta(x)| \, d\mathcal{H}^n(y).$$

By the differentiation property of θ, this converges to 0 as $\lambda \downarrow 0$.
Thus,

$$\int f(y) \, d(\mu_V)_{x,\lambda}(y) \longrightarrow \theta(x) \int_P f \, d\mathcal{H}^n(y)$$

for every $f \in C_c(\mathbb{R}^{n+k})$. Hence $(\mu_V)_{x,\lambda} \rightharpoonup \theta(x) \, \mathcal{H}^n \llcorner P$.
Therefore the approximate tangent space $T_x V$ exists and equals $T_x M$ with multiplicity $\theta(x)$. Since this holds for $\mathcal{H}^n$-almost every $x \in M$, the proof is complete. $\square$

Just as in the case of countably n-rectifiable sets, the converse direction also holds. We state it here for completeness, although we shall not require it in what follows.

Theorem 1.36. *Let $M \subset \mathbb{R}^{n+k}$ be an $\mathcal{H}^n$-measurable set, and let $\theta : M \to (0, \infty)$ be an $\mathcal{H}^n$-measurable function such that*

$$\int_{M \cap K} \theta \, d\mathcal{H}^n < \infty$$

for every compact set $K \subset \mathbb{R}^{n+k}$.
Suppose that for $\mathcal{H}^n$-almost every $x \in M$, there exists an n-dimensional subspace $P_x \subset \mathbb{R}^{n+k}$ such that

$$(\theta \ H^n \llcorner M)_{x,\lambda} \rightharpoonup \theta(x) \, \mathcal{H}^n \llcorner P_x \qquad \text{as } \lambda \downarrow 0.$$

Then M is countably n-rectifiable, and consequently

$$V = v(M, \theta)$$

defines a rectifiable n-varifold in U.

We conclude this section with a final example illustrating that the support of a varifold can be quite strange. The reason we discuss the below example is because in many of the theorems, the object one might be interested in is $\operatorname{spt} V$ (such as in the final Allard regularity theorem), and not M. However, under most "nice" examples, it's reasonable to think of this $\operatorname{spt} V$ as M.

Example 1.37 (Support of a varifold can be strange)**.** The support of a varifold need not coincide with the underlying rectifiable set in any simple way.
Let $\{q_i\}_{i=1}^{\infty}$ be an enumeration of $\mathbb{Q}^2 \subset \mathbb{R}^2$. For each i, let L_i be a horizontal line segment centered at q_i of length 2^{-i}, and define

$$M = \bigcup_{i=1}^{\infty} L_i.$$

Let $\theta \equiv 1$ on M, and consider the associated varifold $V = v(M, 1)$.
Since the rational points are dense in $\mathbb{R}^2$, every ball in $\mathbb{R}^2$ intersects infinitely many of the segments L_i, and hence has positive μ_V-measure. It follows that

$$\operatorname{spt} V = \mathbb{R}^2.$$

Thus, even though M is a very sparse set, the support of the associated varifold is the entire ambient space.

2 Towards the Allard Regularity Theorem

2.1 Extending Familiar Objects in Smooth Case to Varifolds

Throughout the remainder of this report, let $U \subset \mathbb{R}^{n+k}$ be an open set, and let $V = v(M, \theta)$ be a rectifiable n-varifold in U. Thus $M \subset U$ is a countably n-rectifiable, $\mathcal{H}^n$-measurable set, and

$$\mu_V = \theta\, \mathcal{H}^n \llcorner M$$

is a Radon measure on U. We also adpot the convention of thinking of θ as a function on U by defining it to be 0 outside M.

Before extending familiar geometric quantities such as gradients, divergences, and Jacobians to the setting of rectifiable varifolds, let us recall a fundamental fact established in the previous section. Namely, since M is countably n-rectifiable, there exist C^1 n-dimensional submanifolds $\{N_j\}_{j=1}^{\infty}$ of $\mathbb{R}^{n+k}$, together with an $\mathcal{H}^n$-measurable set N_0 satisfying

$$\mathcal{H}^n(N_0) = 0,$$

such that

$$M \subset N_0 \cup \bigcup_{j=1}^{\infty} N_j.$$

Moreover, we proved that for $\mathcal{H}^n$-almost every $x \in M$, the approximate tangent space exists and is independent of the choice of decomposition into C^1 submanifolds. More precisely, whenever $x \in M \cap N_j$,

$$T_x V = T_x M = T_x N_j$$

for $\mathcal{H}^n$-almost every $x \in M \cap N_j$, where:

- $T_x V$ denotes the approximate tangent space of the varifold V at x (with multiplicity $\theta(x)$),
- $T_x M$ denotes the approximate tangent space of the rectifiable set M,
- $T_x N_j$ denotes the usual classical tangent space of the C^1-submanifold N_j.

This fact is the key mechanism which allows one to define differential-geometric quantities on rectifiable varifolds by reducing locally to the C^1 setting.

Definition 2.1 (Gradient and Differential)**.** Let $f : U \to \mathbb{R}$ be a Lipschitz function. Since each $N_j \cap U$ is a C^1-submanifold of $\mathbb{R}^{n+k}$, the tangential gradient $\nabla^{N_j} f(x)$ is well-defined for $\mathcal{H}^n$-almost every $x \in N_j \cap U$.
For $\mathcal{H}^n$-almost every $x \in M$, choose j such that $x \in N_j$, and define

$$\nabla^M f(x) := \nabla^{N_j} f(x).$$

This definition is independent of the choice of N_j, since for $\mathcal{H}^n$-almost every $x \in M$,

$$T_x M = T_x N_j,$$

and the tangential gradient depends only on the tangent space.

We then define the tangential differential

$$df|_x : T_x M \to \mathbb{R}$$

by

$$df|_x(v) = \langle \nabla^M f(x), v \rangle, \qquad v \in T_x M.$$

The preceding construction shows that many familiar differential-geometric objects from the smooth setting extend naturally to rectifiable varifolds. In particular, given a Lipschitz map $f : U \to \mathbb{R}^m$, one may define the tangential Jacobian $J^M f(x)$ for $\mathcal{H}^n$-almost every $x \in M$ by computing the Jacobian of the restriction of f to the approximate tangent space $T_x M$. Similarly, if $X : U \to \mathbb{R}^{n+k}$ is a Lipschitz vector field, one defines the tangential divergence $\operatorname{div}_M X(x)$ by taking the trace of the tangential differential of X along $T_x M$.

We now observe that the area formula continues to hold in this setting. This follows essentially by reducing to the classical area formula on the C^1-submanifolds appearing in a rectifiable decomposition of M.

Theorem 2.2 (Area Formula for Rectifiable Varifolds)**.** *Let $f : U \to \mathbb{R}^m$ be a Lipschitz map with $m \geq n$, and let $A \subset M$ be an $\mathcal{H}^n$-measurable set. Then*

$$\int_A J^M f(x)\, d\mathcal{H}^n(x) = \int_{\mathbb{R}^m} \mathcal{H}^0\Big(f^{-1}(y) \cap A\Big)\, d\mathcal{H}^n(y).$$

More generally, if $g : M \to \mathbb{R}$ is an $L^1_{\mathrm{loc}}(\mathcal{H}^n \llcorner M)$ function, then

$$\int_A g(x)\, J^M f(x)\, d\mathcal{H}^n(x) = \int_{\mathbb{R}^m} \left(\sum_{x \in f^{-1}(y) \cap A} g(x) \right) d\mathcal{H}^n(y).$$

Proof. Since M is countably n-rectifiable, there exist C^1 n-dimensional submanifolds $\{N_j\}_{j=1}^{\infty}$ such that

$$\mathcal{H}^n\left(M \setminus \bigcup_{j=1}^{\infty} N_j \right) = 0.$$

Define

$$A_j = A \cap N_j \setminus \bigcup_{i<j} N_i.$$

Then the sets A_j are pairwise disjoint, $\mathcal{H}^n$-measurable, and $A = \bigcup_{j=1}^{\infty} A_j$ up to an $\mathcal{H}^n$-null set.

Since N_j is a C^1-submanifold, the classical area formula applies on N_j:

$$\int_{A_j} J^{N_j} f(x)\, d\mathcal{H}^n(x) = \int_{\mathbb{R}^m} \mathcal{H}^0\Big(f^{-1}(y) \cap A_j\Big)\, d\mathcal{H}^n(y).$$

Moreover, $J^{N_j} f(x) = J^M f(x)$ for $\mathcal{H}^n$-almost every $x \in A_j$. Summing over j, and using monotone convergence,

$$\int_A J^M f(x)\, d\mathcal{H}^n(x) = \sum_{j=1}^{\infty} \int_{A_j} J^{N_j} f(x)\, d\mathcal{H}^n(x)$$

$$= \sum_{j=1}^{\infty} \int_{\mathbb{R}^m} \mathcal{H}^0\Big(f^{-1}(y) \cap A_j\Big)\, d\mathcal{H}^n(y).$$

Since the A_j are disjoint,

$$\sum_{j=1}^{\infty} \mathcal{H}^0\Big(f^{-1}(y) \cap A_j\Big) = \mathcal{H}^0\Big(f^{-1}(y) \cap A\Big),$$

and hence

$$\int_A J^M f(x)\, d\mathcal{H}^n(x) = \int_{\mathbb{R}^m} \mathcal{H}^0\Big(f^{-1}(y) \cap A\Big)\, d\mathcal{H}^n(y).$$

The second statement follows similarly by applying the weighted area formula on each N_j and then using the monotone convergence theorem. □

2.2 First Variation Formula for Rectifiable Varifolds

Definition 2.3 (Pushforward of a Varifold)**.** Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$, and let $W \subset \mathbb{R}^m$ be open. Suppose $f : U \to W$ is a Lipschitz map which is proper, i.e. the preimage of every compact subset of W is compact in M. We define the pushforward varifold $f_\# V$ by

$$f_\# V := v(f(M), \widetilde{\theta}),$$

where the multiplicity function $\widetilde{\theta}$ is given by

$$\widetilde{\theta}(y) = \sum_{x \in f^{-1}(y) \cap M} \theta(x).$$

We now verify that $\widetilde{\theta}$ is locally integrable with respect to $\mathcal{H}^n \llcorner f(M)$, so that $f_\# V$ is indeed a rectifiable varifold.

Let $K \subset W$ be compact. Since f is proper, $f^{-1}(K) \cap M$ is compact in M. Applying the area formula to f, we obtain

$$\int_{K \cap f(M)} \widetilde{\theta}(y)\, d\mathcal{H}^n(y) = \int_{f^{-1}(K) \cap M} \theta(x)\, J^M f(x)\, d\mathcal{H}^n(x).$$

Since f is Lipschitz, the tangential Jacobian $J^M f$ is locally bounded. Hence,

$$\int_{f^{-1}(K)\cap M} \theta(x)\, J^M f(x)\, d\mathcal{H}^n(x) < \infty,$$

because $\theta\, \mathcal{H}^n \llcorner M$ is a Radon measure. Thus,

$$\widetilde{\theta} \in L^1_{\mathrm{loc}}(\mathcal{H}^n \llcorner f(M)).$$

Remark 2.4. If f is injective, then for $y \in f(M)$, $\widetilde{\theta}(y) = \theta(f^{-1}(y))$.

The mass of the pushforward varifold is given by

$$\mathbf{M}(f_{\#}V) = \int_{f(M)} \widetilde{\theta}\, d\mathcal{H}^n = \int_M J^M f(x)\, d\mu_V(x),$$

or equivalently,

$$\mathbf{M}(f_{\#}V) = \int_M \theta(x)\, J^M f(x)\, d\mathcal{H}^n(x).$$

We can now derive an analogous first variation formula of area for rectifiable varfiolds. We first recall the definition of a perturbation (with a slight modification so that it makes sense for varifolds):

Definition 2.5 (Perturbations of a submanifold)**.** Let $V = v(M, \theta)$ be a rectfiable varifold in $U \subset \mathbb{R}^{n+k}$.
A map $\Phi : (-\varepsilon, \varepsilon) \times U \longrightarrow \mathbb{R}^{n+k}$ is called a *perturbation* (or *variation*) of V if:

(i) $\dfrac{\partial \Phi}{\partial t} : (-\varepsilon, \varepsilon) \times U \to \mathbb{R}^{n+k}$ is of class C^1,

(ii) $\Phi(0, x) = x$ for all $x \in U$ and $\Phi(t, \cdot)$ is injective for each t.

The perturbation is said to be *compactly supported* (or a *compact perturbation*) if there exists a compact set $K \subset U$ such that

$$\Phi(t, x) = x \qquad \text{for all } x \in U \setminus K \text{ and all } t \in (-\varepsilon, \varepsilon).$$

Theorem 2.6 (First Variation Formula for Rectifiable Varifolds)**.** *Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$, and let*

$$\Phi : (-\varepsilon, \varepsilon) \times U \to \mathbb{R}^{n+k}$$

be a compactly supported perturbation in the sense of Definition 2.5. For each t, define $\Phi_t(x) := \Phi(t, x)$, and let

$$X(x) := \frac{\partial \Phi}{\partial t}(t, x)\bigg|_{t=0}.$$

Then

$$\frac{d}{dt}\bigg|_{t=0} \mathbf{M}((\Phi_t)_{\#}V) = \int_M \operatorname{div}_M X(x)\, d\mu_V(x),$$

or equivalently,

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \theta(x) \operatorname{div}_M X(x) \, d\mathcal{H}^n(x).$$

Proof. Since Φ_t is Lipschitz and proper on $\operatorname{spt} V$, the pushforward varifold $(\Phi_t)_\# V$ is well-defined. By the area formula established previously,

$$\mathbf{M}((\Phi_t)_\# V) = \int_M J^M \Phi_t(x) \, d\mu_V(x).$$

Thus,

$$\frac{\mathbf{M}((\Phi_t)_\# V) - \mathbf{M}(V)}{t} = \int_M \frac{J^M \Phi_t(x) - 1}{t} \, d\mu_V(x).$$

From the first variation formula for C^2 submanifolds, we already know that

$$\left.\frac{d}{dt}\right|_{t=0} J^M \Phi_t(x) = \operatorname{div}_M X(x),$$

and moreover this convergence is uniform on compact subsets of M.

Since the perturbation is compactly supported, there exists a compact set $K \subset U$ such that

$$\Phi_t(x) = x \qquad \text{for all } x \notin K.$$

Hence,

$$J^M \Phi_t(x) = 1 \qquad \text{for all } x \notin K,$$

so the above integrand vanishes outside K.

Therefore, by dominated convergence,

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \left.\frac{d}{dt}\right|_{t=0} J^M \Phi_t(x) \, d\mu_V(x),$$

which gives

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \operatorname{div}_M X(x) \, d\mu_V(x).$$

Finally, since

$$d\mu_V = \theta \, d\mathcal{H}^n \llcorner M,$$

we may equivalently write

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \theta(x) \operatorname{div}_M X(x) \, d\mathcal{H}^n(x).$$

□

Definition 2.7 (Stationary Varifold)**.** Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$. We say that V is *stationary* in U if for every compactly supported perturbation $\Phi : (-\varepsilon, \varepsilon) \times U \to \mathbb{R}^{n+k}$, one has

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = 0,$$

Equivalently, by the first variation formula, V is stationary if

$$\int_M \operatorname{div}_M X(x)\, d\mu_V(x) = 0$$

for every compactly supported C^1 vector field

$$X : U \to \mathbb{R}^{n+k}.$$

2.3 Rectifiable Varifolds in Ambient Submanifold and Generalized Mean Curvature

We now discuss rectifiable varifolds in a slightly more general setting. Namely, instead of working in an open subset of $\mathbb{R}^{n+k}$, we consider varifolds contained in a fixed embedded submanifold

$$N^{n+\ell} \subset \mathbb{R}^{n+k}, \qquad \ell \leq k.$$

All of the preceding notions extend naturally to this setting, and in particular one obtains an analogous first variation formula relative to the ambient manifold N.

Definition 2.8 (Rectifiable Varifold in a Submanifold)**.** Let $N^{n+\ell} \subset \mathbb{R}^{n+k}$ be an embedded C^1 submanifold. Let $M \subset N$ be a countably n-rectifiable, $\mathcal{H}^n$-measurable set, and let $\theta : M \to (0, \infty)$ be $\mathcal{H}^n$-measurable such that

$$\int_{M \cap K} \theta \, d\mathcal{H}^n < \infty$$

for every compact set $K \subset N$.
Then $V = v(M, \theta)$ is called a rectifiable n-varifold in N.

Observe that every such varifold may also be viewed as a rectifiable varifold in a suitable open subset of $\mathbb{R}^{n+k}$. Indeed, for each $x \in N$, since N is embedded, there exists $r_x > 0$ such that $\overline{B(x, r_x)} \cap N$ is compact in N. Define

$$U := \bigcup_{x \in N} B(x, r_x),$$

which is an open subset of $\mathbb{R}^{n+k}$ containing N.
Now let $K \subset U$ be compact. Since $\{B(x, r_x)\}_{x \in N}$ covers K, compactness yields a finite subcover

$$K \subset \bigcup_{i=1}^{m} B(x_i, r_i).$$

Hence,

$$K \cap N \subset \bigcup_{i=1}^{m} \left(\overline{B(x_i, r_i)} \cap N \right).$$

The right-hand side is compact in N, and since $K \cap N$ is closed in it, $K \cap N$ is compact in N. Therefore,

$$\int_{M \cap K} \theta \, d\mathcal{H}^n = \int_{M \cap (K \cap N)} \theta \, d\mathcal{H}^n < \infty,$$

showing that V may indeed be regarded as a rectifiable varifold in $U \subset \mathbb{R}^{n+k}$.

Definition 2.9 (Perturbations in a Submanifold)**.** Let $V = v(M, \theta)$ be a rectifiable varifold in N.
A map $\Phi : (-\varepsilon, \varepsilon) \times N \to N$ is called a *perturbation* (or *variation*) of V in N if:

(i) $\frac{\partial \Phi}{\partial t} : (-\varepsilon, \varepsilon) \times N \to \mathbb{R}^{n+k}$ is of class C^1,

(ii) $\Phi(0, x) = x \qquad$ for all $x \in N$, and for each t, the map $\Phi_t := \Phi(t, \cdot)$ is injective.

The perturbation is said to be *compactly supported* if there exists a compact set $K \subset N$ such that

$$\Phi(t, x) = x \qquad \text{for all } x \in N \setminus K \text{ and all } t \in (-\varepsilon, \varepsilon).$$

Definition 2.10 (Stationary Varifold in a Submanifold)**.** Let $V = v(M, \theta)$ be a rectifiable varifold in N.
We say that V is *stationary in N* if

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}\Big((\Phi_t)_\# V\Big) = 0$$

for every compactly supported perturbation $\Phi : (-\varepsilon, \varepsilon) \times N \to N$.

We now observe that the first variation formula immediately yields an equivalent characterization of stationarity in N.

Proposition 2.11. *Let $V = v(M, \theta)$ be a rectifiable varifold in an embedded submanifold $N^{n+\ell} \subset \mathbb{R}^{n+k}$. Then V is stationary in N if and only if*

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for every compactly supported C^1 vector field $X : N \to \mathbb{R}^{n+k}$ such that $X(x) \in T_x N$ for all $x \in N$.

Proof. Suppose first that V is stationary in N. Let $X : N \to \mathbb{R}^{n+k}$ be a compactly supported C^1 vector field tangent to N. Since $X(x) \in T_x N$ for all x, the flow generated by X produces a compactly supported perturbation $\Phi : (-\varepsilon, \varepsilon) \times N \to N$. Applying the first variation formula gives

$$0 = \left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \operatorname{div}_M X \, d\mu_V.$$

Conversely, suppose

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for every compactly supported tangential vector field X. Let $\Phi : (-\varepsilon, \varepsilon) \times N \to N$ be a compactly supported perturbation, and define

$$X(x) = \left.\frac{\partial \Phi}{\partial t}(t, x)\right|_{t=0}.$$

Since $\Phi_t(N) \subset N$, necessarily $X(x) \in T_x N$ for all $x \in N$. Hence,

$$\left.\frac{d}{dt}\right|_{t=0} \mathbf{M}((\Phi_t)_\# V) = \int_M \operatorname{div}_M X \, d\mu_V = 0.$$

Thus V is stationary in N. $\square$

We now recall the definition of the second fundamental form of an embedded submanifold.

Definition 2.12 (Second Fundamental Form)**.** Let $N^{n+\ell} \subset \mathbb{R}^{n+k}$ be an embedded C^2 submanifold. For $x \in N$, let

$$\{\eta_1(x), \ldots, \eta_{k-\ell}(x)\}$$

be an orthonormal basis of $T_x N^\perp$.

The second fundamental form of N at x is the bilinear map

$$B^N(x) : T_x N \times T_x N \to T_x N^\perp$$

defined by

$$B^N(x)(v, w) = \sum_{i=1}^{k-\ell} \langle v, D_w \eta_i(x) \rangle \, \eta_i(x).$$

If $\{\tau_1, \ldots, \tau_{n+\ell}\}$ is an orthonormal basis of $T_x N$, then the mean curvature vector of N is defined by

$$H^N(x) = \sum_{i=1}^{n+\ell} B^N(x)(\tau_i, \tau_i).$$

Equivalently,

$$H^N(x) = -\sum_{i=1}^{k-\ell} (\operatorname{div}_N \eta_i(x)) \, \eta_i(x).$$

Now let $V = v(M, \theta)$ be a rectifiable n-varifold in N. At points $x \in M$ where the approximate tangent space T_xM exists, define the mean curvature of N along M by

$$H_M^N(x) = \sum_{i=1}^{n} B^N(x)(v_i, v_i),$$

where $\{v_1, \dots, v_n\}$ is an orthonormal basis of T_xM.
Equivalently,

$$H_M^N(x) = -\sum_{i=1}^{k-\ell} (\operatorname{div}_M \eta_i(x))\, \eta_i(x).$$

We can now characterize stationarity in N using arbitrary ambient vector fields.

Theorem 2.13. *Let $V = v(M, \theta)$ be a rectifiable varifold in an embedded submanifold $N^{n+\ell} \subset \mathbb{R}^{n+k}$. Then V is stationary in N if and only if*

$$\int_M \operatorname{div}_M X \, d\mu_V = -\int_M \langle X, H_M^N \rangle \, d\mu_V$$

for every compactly supported C^1 vector field $X : N \to \mathbb{R}^{n+k}$., X is not necessarily tangential to N

Proof. Suppose first that V is stationary in N. Let $X : N \to \mathbb{R}^{n+k}$ be compactly supported and C^1. Decompose $X = X^\top + X^\perp$, where $X^\top(x) \in T_xN$ and $X^\perp(x) \in T_xN^\perp$. Since V is stationary in N,

$$\int_M \operatorname{div}_M X^\top \, d\mu_V = 0.$$

Hence,

$$\int_M \operatorname{div}_M X \, d\mu_V = \int_M \operatorname{div}_M X^\perp \, d\mu_V.$$

Write

$$X^\perp = \sum_{i=1}^{k-\ell} c_i \, \eta_i,$$

where each c_i is a C^1 function on N. Then

$$\operatorname{div}_M X^\perp = \sum_{i=1}^{k-\ell} \operatorname{div}_M(c_i \eta_i).$$

Using the product rule,

$$\operatorname{div}_M(c_i \eta_i) = \langle \nabla^M c_i, \eta_i \rangle + c_i \operatorname{div}_M \eta_i.$$

Since $\nabla^M c_i \in T_xM \subset T_xN$ while $\eta_i \in T_xN^\perp$, $\langle \nabla^M c_i, \eta_i \rangle = 0$. Therefore,

$$\operatorname{div}_M X^\perp = \sum_{i=1}^{k-\ell} c_i \operatorname{div}_M \eta_i.$$

Using the formula for H_M^N,

$$H_M^N = -\sum_{i=1}^{k-\ell} (\operatorname{div}_M \eta_i)\eta_i,$$

we obtain $\operatorname{div}_M X^\perp = -\langle X^\perp, H_M^N\rangle$. Since $H_M^N \in T_xN^\perp$, $\langle X^\top, H_M^N\rangle = 0$, and thus $\operatorname{div}_M X = -\langle X, H_M^N\rangle$. Integrating over M gives the result.

Conversely, suppose

$$\int_M \operatorname{div}_M X \, d\mu_V = -\int_M \langle X, H_M^N\rangle \, d\mu_V$$

for all compactly supported C^1 vector fields X. If X is tangent to N, then $\langle X, H_M^N\rangle = 0$, since $H_M^N \in T_xN^\perp$. Hence,

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for all compactly supported tangential vector fields X, which implies that V is stationary in N. $\square$

The previous theorem motivates an important generalization of the classical notion of mean curvature to the setting of rectifiable varifolds.

Recall that if $M \subset \mathbb{R}^{n+k}$ is an n-dimensional C^1 submanifold with mean curvature vector H, then for every compactly supported C^1 vector field $X : \mathbb{R}^{n+k} \to \mathbb{R}^{n+k}$, we have the first variation formula

$$\int_M \operatorname{div}_M X \, d\mathcal{H}^n = -\int_M \langle X, H\rangle \, d\mathcal{H}^n.$$

For a general rectifiable varifold, however, there is no a priori smooth structure from which a classical mean curvature vector can be defined. This leads to the following weak formulation.

Definition 2.14 (Generalized Mean Curvature)**.** Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$.
We say that V has *generalized mean curvature* $H : U \to \mathbb{R}^{n+k}$ if for every compactly supported C^1 vector field $X : U \to \mathbb{R}^{n+k}$, we have

$$\int_M \operatorname{div}_M X \, d\mu_V = -\int_M \langle X, H\rangle \, d\mu_V.$$

We immediately obtain two important examples.

Remark 2.15. If V is stationary in U, then

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for every compactly supported vector field X. Hence V has generalized mean curvature $H \equiv 0$.

Remark 2.16. Let $N \subset \mathbb{R}^{n+k}$ be an embedded $(n+\ell)$-dimensional C^1 submanifold, where $\ell \leq k$, and let $V = v(M,\theta)$ be stationary in N. From the first variation formula in N, we showed that

$$\int_M \operatorname{div}_M X \, d\mu_V = -\int_M \langle X, H_M^N \rangle \, d\mu_V$$

for every compactly supported C^1 vector field $X : N \to \mathbb{R}^{n+k}$, where H_M^N denotes the mean curvature of N along M. Thus V has generalized mean curvature $H = H_M^N$.

2.4 The Density Theorems and Monotonicity Formulas

We begin this section with an important theorem we get for varifolds given the existence of the approximate tangent space at a point. We begin with this, as we shall revisit in the middle of the section once more having established the monotonincity formulas

Theorem 2.17. *Let $V = v(M,\theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$. Let $x \in M$ be a point at which the approximate tangent space T_xV exists with multiplicity $\theta(x)$. Then*

$$\lim_{\lambda \downarrow 0} \frac{\mu_V(B_\lambda(x))}{\omega_n \lambda^n} = \theta(x),$$

where $\omega_n = \mathcal{L}^n(B_1(0))$ denotes the volume of the unit ball in $\mathbb{R}^n$.
In particular, if $\theta \equiv 1$, then

$$\lim_{\lambda \downarrow 0} \frac{\mathcal{H}^n(M \cap B_\lambda(x))}{\omega_n \lambda^n} = 1$$

for $\mathcal{H}^n$-almost every $x \in M$.

Proof. Since T_xV exists with multiplicity $\theta(x)$, we know that for every $f \in C_c(\mathbb{R}^{n+k})$,

$$\lim_{\lambda \downarrow 0} \int_{\frac{M-x}{\lambda}} f(y)\, \theta(x+\lambda y)\, d\mathcal{H}^n(y) = \theta(x) \int_{T_xM} f(y)\, d\mathcal{H}^n(y).$$

Fix $\varepsilon > 0$. Choose a continuous function $f_\varepsilon \in C_c(\mathbb{R}^{n+k})$ such that $0 \leq f_\varepsilon \leq 1$, $f_\varepsilon \equiv 1 \quad$ on $B_1(0)$, and $f_\varepsilon \equiv 0 \quad$ outside $B_{1+\varepsilon}(0)$.

Applying the above convergence to f_ε, we obtain

$$\lim_{\lambda \downarrow 0} \int_{\frac{M-x}{\lambda}} f_\varepsilon(y)\, \theta(x+\lambda y)\, d\mathcal{H}^n(y) = \theta(x) \int_{T_xM} f_\varepsilon(y)\, d\mathcal{H}^n(y).$$

Now,

$$\int_{\frac{M-x}{\lambda}} f_\varepsilon(y)\, \theta(x+\lambda y)\, d\mathcal{H}^n(y) = \frac{1}{\lambda^n} \int_M f_\varepsilon\Big(\frac{y-x}{\lambda}\Big)\, \theta(y)\, d\mathcal{H}^n(y).$$

Since $f_\varepsilon \equiv 1$ on $B_1(0)$ and vanishes outside $B_{1+\varepsilon}(0)$,

$$\frac{\mu_V(B_\lambda(x))}{\lambda^n} \leq \frac{1}{\lambda^n} \int_M f_\varepsilon\Big(\frac{y-x}{\lambda}\Big)\, d\mu_V(y) \leq \frac{\mu_V(B_{(1+\varepsilon)\lambda}(x))}{\lambda^n}.$$

Passing to the limit as $\lambda \downarrow 0$, we obtain

$$\theta(x)\omega_n \leq \liminf_{\lambda \downarrow 0} \frac{\mu_V(B_{(1+\epsilon)\lambda}(x))}{\lambda^n}$$

and

$$\limsup_{\lambda \downarrow 0} \frac{\mu_V(B_\lambda(x))}{\lambda^n} \leq \theta(x)\, \mathcal{H}^n\Big(T_x M \cap B_{1+\varepsilon}(0)\Big).$$

Since $T_x M$ is an n-dimensional plane,

$$\mathcal{H}^n(T_x M \cap B_{1+\varepsilon}(0)) = \omega_n (1+\varepsilon)^n.$$

Letting $\varepsilon \downarrow 0$ yields

$$\lim_{\lambda \downarrow 0} \frac{\mu_V(B_\lambda(x))}{\omega_n \lambda^n} = \theta(x).$$

The final statement follows immediately in the case $\theta \equiv 1$. □

We can now move onto what is probably one of the most important results for stationary varifolds, the monotonicity formulas. The proof however, unlike the importance of the result, is a mere, albeit tedious calculation which we omit. We shall however make a few observations as corollaries to this theorem.

Theorem 2.18 (Monotonicity Formula). *Let $V = v(M, \theta)$ be a stationary rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$, and let $\zeta \in U$. Suppose $0 < \sigma_1 \leq \sigma_2 < R$, where $R > 0$ is such that $B_R(\zeta) \subset U$. Then*

$$\frac{\mu_V(B_{\sigma_2}(\zeta))}{\omega_n \sigma_2^n} - \frac{\mu_V(B_{\sigma_1}(\zeta))}{\omega_n \sigma_1^n} = \int_{B_{\sigma_2}(\zeta) \setminus B_{\sigma_1}(\zeta)} \frac{|P_{(T_x V)^\perp}(x-\zeta)|^2}{\omega_n |x-\zeta|^{n+2}}\, d\mu_V(x),$$

where $P_{(T_x V)^\perp}$ denotes the orthogonal projection onto the orthogonal complement of the approximate tangent space $T_x V$.

In particular, the function

$$\sigma \longmapsto \frac{\mu_V(B_\sigma(\zeta))}{\omega_n \sigma^n}$$

is monotone increasing in σ.

Proof. We omit the details of the proof, as it consists primarily of a direct, though somewhat lengthy, computation.
The main idea is the following. Since V is stationary, we have

$$\int_M \operatorname{div}_M X \, d\mu_V = 0$$

for every compactly supported C^1 vector field $X : U \to \mathbb{R}^{n+k}$.
One then chooses a radial vector field centered at ζ of the form

$$X(x) = \gamma(r)(x-\zeta), \qquad r = |x-\zeta|,$$

where γ is a suitable smooth cutoff function. Substituting this vector field into the stationarity identity and computing $\operatorname{div}_M X$ yields, after rearrangement, the stated formula. $\square$

Definition 2.19 (Cones and Local Cones)**.** Let $A \subset \mathbb{R}^{n+k}$, and let $\zeta \in \mathbb{R}^{n+k}$.
We say that A is a *cone around* ζ if whenever $x \in A$,

$$\zeta + t(x - \zeta) \in A \qquad \text{for all } t > 0.$$

We say that A is a *local cone around* ζ if there exists $R > 0$ such that whenever $x \in A \cap B_R(\zeta)$, we have

$$\zeta + t(x - \zeta) \in A \cap B_R(\zeta)$$

for all $0 \leq t < \frac{R}{|x-\zeta|}$.

We would now like to reformulate the above definition in a way that makes sense for rectifiable varifolds. Since varifolds are defined only up to $\mathcal{H}^n$-null sets, and also contains a mulitplicity function we need to account for, the previous formulation in terms of points of a set is not entirely appropriate, although it captures the correct geometric intuition. For $\zeta \in \mathbb{R}^{n+k}$ and $c > 0$, define the dilation map

$$\tau_{\zeta,c}(x) = \zeta + c(x - \zeta).$$

Definition 2.20 (Local Cone Varifold)**.** Let $V = v(M, \theta)$ be a rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$.. We say that V is *locally a cone around* ζ if there exists $R > 0$ such that

$$(\tau_{\zeta,t})_{\#}\Big(V \llcorner B_{\frac{R}{t}}(\zeta)\Big) = V \llcorner B_R(\zeta)$$

for every $0 < t < R$.

Intuitively, this definition says that after rescaling V about the point ζ, the varifold remains unchanged. Up to $\mathcal{H}^n$-null sets, this is equivalent to the following: If $x \in M \cap B_R(\zeta)$, and $r = |x - \zeta|$, then for every $0 \leq t < \frac{R}{r}$, the point $x \in B(x, \frac{R}{t})$, and thus $\zeta + t(x - \zeta)$ also lies in $M \cap B_R(\zeta)$, which is precisely the geometric scaling property of a cone. Moreover, the multiplictiy function θ also satisfies a similar invariance under scaling, namely that $\theta(\zeta + t(x - \zeta) = \theta(x)$ for all $0 < t < \frac{R}{|x-\zeta|}$. In other words, θ is a radial inside the conical region.

Theorem 2.21. *Let $V = v(M, \theta)$ be a stationary rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$, and let $\zeta \in U$. Suppose there exists $R > 0$ such that $B_R(\zeta) \subset U$ and such that the density ratio*

$$\sigma \longmapsto \frac{\mu_V(B_\sigma(\zeta))}{\omega_n \sigma^n}$$

is constant for all $0 < \sigma < R$.

Then V is locally a cone around ζ. More precisely,

$$(\tau_{\zeta,t/R})_{\#}\Big(V \llcorner B_R(\zeta)\Big) = V \llcorner B_t(\zeta)$$

for every $0 < t < R$.

Proof. We only sketch the main idea of the proof, since the complete argument is somewhat involved. The essential geometric content already appears in the following computation.

Since the density ratio is constant, the monotonicity formula implies that

$$\int_{B_{\sigma_2}(\zeta)\setminus B_{\sigma_1}(\zeta)} \frac{|P_{(T_xV)^\perp}(x-\zeta)|^2}{|x-\zeta|^{n+2}}\, d\mu_V(x) = 0$$

for every $0 < \sigma_1 < \sigma_2 < R$.

Hence,

$$P_{(T_xV)^\perp}(x-\zeta) = 0$$

for μ_V-almost every $x \in B_R(\zeta)$.

Equivalently,

$$x - \zeta \in T_xM$$

for $\mathcal{H}^n$-almost every $x \in M \cap B_R(\zeta)$. Thus, up to $\mathcal{H}^n$-null sets, the vector field

$$X(x) = x - \zeta$$

is tangential to M.

The flow associated to this vector field is obtained by solving

$$\frac{d}{dt}\Phi_t(x) = \Phi_t(x) - \zeta, \qquad \Phi_0(x) = x,$$

which yields

$$\Phi_t(x) = \zeta + e^t(x-\zeta).$$

Since the vector field is tangential to M, its flow preserves M (up to $\mathcal{H}^n$-null sets). Therefore, whenever

$$x \in M \cap B_R(\zeta),$$

we have

$$\zeta + s(x-\zeta) \in M$$

for all

$$0 < s < \frac{R}{|x-\zeta|}.$$

Thus M is locally conical around ζ in the sense described earlier. The remaining step, namely proving that the multiplicity function θ is constant along rays emanating from ζ, is slightly more delicate and will be omitted. □

Theorem 2.22 (Existence of Density for Stationary Varifolds)**.** *Let $V = v(M,\theta)$ be a stationary rectifiable n-varifold in an open set $U \subset \mathbb{R}^{n+k}$. Then for every $\zeta \in U$, the density*

$$\Theta^n(\mu_V, \zeta) := \lim_{\rho \downarrow 0} \frac{\mu_V(B_\rho(\zeta))}{\omega_n \rho^n}$$

exists.
Moreover, $\Theta^n(\mu_V, x) = \theta(x)$ for $\mathcal{H}^n$-almost every $x \in M$.

Proof. By the monotonicity formula for stationary varifolds established previously,

$$\rho \longmapsto \frac{\mu_V(B_\rho(\zeta))}{\omega_n \rho^n}$$

is monotone increasing for every $\zeta \in U$. Hence the limit

$$\lim_{\rho \downarrow 0} \frac{\mu_V(B_\rho(\zeta))}{\omega_n \rho^n}$$

exists for every $\zeta \in U$, proving existence of the density.

Now let $x \in M$ be such that the approximate tangent space $T_x V$ exists with multiplicity $\theta(x)$. By the density theorem proved earlier,

$$\lim_{\rho \downarrow 0} \frac{\mu_V(B_\rho(x))}{\omega_n \rho^n} = \theta(x).$$

Since approximate tangent spaces exist for $\mathcal{H}^n$-almost every point of M, the conclusion follows. □

We conclude this section by stating analogous monotonicity and conical approximation results in the more general setting of rectifiable varifolds with L^p generalized mean curvature. Observe that the stationary case considered previously corresponds precisely to the special case in which the generalized mean curvature vanishes identically.

Theorem 2.23 (Monotonicity Formula under an L^p Bound on the Generalized Mean Curvature)**.** *Let $U \subset \mathbb{R}^{n+k}$ be open, and let $V = v(M,\theta)$ be an n-dimensional rectifiable varifold in U with generalized mean curvature H.*
Fix $\zeta \in U$ and $R > 0$ such that $B_R(\zeta) \subset U$. Assume that $p > n$ and that

$$R^{p-n} \left(\int_{B_R(\zeta)} |H|^p \, d\mu_V \right)^{1/p} \leq \Gamma$$

for some constant $\Gamma \geq 0$.
Then there exist functions $F : (0,R) \to \mathbb{R}$, $G : (0,R) \times (0,R) \to (0,\infty)$, satisfying

$$|F(\rho)| \leq \Gamma \left(\frac{\rho}{R} \right)^{1-\frac{n}{p}} \quad \text{and} \quad e^{-\Gamma} \leq G(\sigma,\rho) \leq e^{\Gamma}$$

for all $0 < \sigma < \rho < R$, such that

$$e^{F(\rho)}\frac{\mu_V(B_\rho(\zeta))}{\omega_n\rho^n} + \frac{1}{p}(e^{F(\rho)}-1) - e^{F(\sigma)}\frac{\mu_V(B_\sigma(\zeta))}{\omega_n\sigma^n} - \frac{1}{p}(e^{F(\sigma)}-1)$$
$$= \frac{1}{\omega_n}\int_{B_\rho(\zeta)\setminus B_\sigma(\zeta)} G(\sigma,\rho)\frac{|P_{(T_xV)^\perp}(x-\zeta)|^2}{|x-\zeta|^{n+2}}\,d\mu_V(x),$$

where $P_{(T_xV)^\perp}$ denotes orthogonal projection onto the orthogonal complement of the approximate tangent space T_xV.
In particular, the quantity

$$e^{F(\rho)}\frac{\mu_V(B_\rho(\zeta))}{\omega_n\rho^n} + \frac{1}{p}\Big(e^{F(\rho)}-1\Big)$$

is monotone increasing in $\rho \in (0,R)$.

The following theorem gives a quantitative version of the fact that if the generalized mean curvature is sufficiently small and the density ratios are almost constant, then the varifold is close to being conical. It is a generalization of the conical result we considered earlier for constant density in the stationary case.

Theorem 2.24 (Conical Approximation)**.** *Suppose $\Lambda > 0$, $p > n$, $\varepsilon \in \Big(0,\frac{1}{4}\Big]$, $\delta \in \Big[0,\frac{1}{16}\Big]$, and let $V = v(M,\theta)$ be a rectifiable n-varifold with generalized mean curvature H in a neighborhood of $B_1(0)$.*
Assume that $0 \in \operatorname{spt} V$, $\Theta^n(\mu_V,0) \geq 1$, $\frac{\mu_V(B_1(0))}{\omega_n} \leq \Theta^n(\mu_V,0)+\delta$, and $\Big(\int_{B_1(0)}|H|^p\,d\mu_V\Big)^{1/p} \leq \delta$.
Then there exists a constant $C = C(n,p,\varepsilon)$ such that

$$\frac{\mu_V(B_\rho(\zeta))}{\omega_n\rho^n} \leq \Theta^n(\mu_V,0) + C(\delta+\varepsilon)$$

for all $\zeta \in B_{1/2}(0)\setminus B_{2\varepsilon}(0)$, all $\rho \in [\varepsilon,1]$, and all admissible scales satisfying

$$\rho < \min\{|\zeta|, (1+\delta/4)-|\zeta|\}\,.$$

Roughly speaking, the theorem says that if the generalized mean curvature is sufficiently small and the mass ratios are close to constant, then the varifold behaves approximately like a cone at sufficiently small scales.

2.5 The Statement of the Allard Regularity Theorem

We begin by stating a baby version of the Allard Regularity Theorem, albeit an extremely important one. It is precisely this result that much of the preceding theory was designed to work towards. Roughly speaking, the theorem below states that if a stationary rectifiable varifold has multiplicity function $\theta \equiv 1$, so that we may heuristically think of the varifold simply as the underlying set itself, then the surface must in fact be locally a smooth minimal submanifold for $\mathcal{H}^n$-almost every point.

This is already a remarkable phenomenon: a priori, rectifiable varifolds are extremely weak geometric objects defined only in a measure-theoretic sense, with no assumed smoothness whatsoever. Nevertheless, stationarity alone forces a strong amount of hidden regularity, upgrading this weak notion of surface to a smooth minimal submanifold almost everywhere.

Theorem 2.25 (Baby Allard Regularity Theorem I). *Let $V = v(M, \theta)$ be an n-dimensional rectifiable varifold in an open set $U \subset \mathbb{R}^{n+k}$. Assume that:*

(i) V is stationary in U;

(ii) $\theta(x) \equiv 1$ for $\mathcal{H}^n$-a.e. $x \in M$.

Then for $\mathcal{H}^n$-almost every point $x \in M$, there exists $r_x > 0$ such that, after a orthogonal transformation, $\operatorname{spt} V \cap B_{r_x}(x)$ is the graph of a smooth function. In particular, M is locally a smooth embedded minimal submanifold near $\mathcal{H}^n$-almost every point.

Next, we state a similar but slightly stronger version of the previous result. Once again, we assume that $V = v(M, \theta)$ is a stationary rectifiable varifold in an open set U, but this time we only assume that the multiplicity satisfies $\theta \geq 1$. Thus, the varifold may now consist of several sheets stacked together, and singular behaviour is no longer ruled out a priori.

Nevertheless, the following version of the Allard Regularity Theorem says that at points where the mass density remains sufficiently close to 1, the varifold still behaves like a single smooth sheet. More precisely, if the density ratio

$$\frac{\mu_V(B_\rho(x))}{\omega_n \rho^n}$$

does not deviate too much from 1 at some scale, then locally around x the support of the varifold is given, after a suitable rotation, by the graph of a C^∞ function. In particular, near such points the varifold is locally a smooth minimal submanifold.

Theorem 2.26 (Baby Allard Regularity Theorem II). *Let $V = v(M, \theta)$ be an n-dimensional rectifiable varifold in an open set $U \subset \mathbb{R}^{n+k}$.. Then there exists a constant $\delta_0 = \delta_0(n, k) > 0$ with the following property:*
If the varifold satisfies the following:

(i) V is stationary in U;

(ii) $\theta(x) \geq 1$ for $\mathcal{H}^n$-a.e. $x \in M$.

(iii) Let $x \in \operatorname{spt} V$ and there exists $\rho > 0$ such that

$$\frac{\mu_V(B_\rho(x))}{\omega_n \rho^n} < 1 + \delta_0.$$

Then there exists $0 < \rho' < \rho$ such that, after a suitable orthogonal transformation of $\mathbb{R}^{n+k}$, $\operatorname{spt} V \cap B_{\rho'}(x)$ is the graph of a C^∞ function.

We conclude this report with the full statement of the Allard Regularity Theorem, the most general version. In contrast with the previous results, we now allow the varifold to have nonzero generalized mean curvature, but assume that this curvature is sufficiently small in an appropriate L^p sense near a given point. We continue to assume that the multiplicity satisfies $\theta \geq 1$ and the mass ratio around the given point is sufficient small at a small scale.

Then Allard Regularity Theorem then states that, after a suitable orthogonal transformation, the support of the varifold near the origin is given by the graph of a $C^{1,\alpha}$ function. In particular, despite the fact that the varifold is initially defined only in a weak measure-theoretic sense, sufficiently small generalized mean curvature together with near-flat density behaviour forces strong local regularity. This is made precisely in the following:

Theorem 2.27 (Allard Regularity Theorem). *Let $V = v(M, \theta)$ be an n-dimensional rectifiable varifold in an open set $U \subset \mathbb{R}^{n+k}$, s.t. $\theta(y) \geq 1$ for μ_V a.e. y. Let $p > n$. Then there are constant $\delta_0 = \delta_0(p, n, k)$ and $\gamma_0 = \gamma_0(p, n, k)$ s.t. the following holds:*
If:

(i) $x \in \operatorname{spt} V$ and $B_\rho(x) \subset U$;

(ii) $\mu_V(B_\rho(x)) \leq (1 + \delta)\omega_n \rho^n$ where $\delta < \delta_0$

(iii) $p > n$ and $\left(\rho^{p-n} \int_{B_\rho(x)} |H|^p \, d\mu_V\right)^{1/p} \leq \delta$,

Then there exist constants $C = C(n, k, p) > 0$, $\alpha = 1 - \frac{n}{p}$, an orthogonal transformation $Q \in O(n + k)$, and a function $u = (u^1, \ldots, u^k) \in C^{1,\alpha}(B^n_{\rho\gamma_0}(x), \mathbb{R}^k)$ such that:

(a) $Du(x) = 0; \operatorname{spt} V \cap B_{\rho\gamma_0}(x) = Q(\operatorname{graph} u) \cap B_{\rho\gamma_0}(x)$;

(b)

$$\sigma^{-1} \sup_{B^n_{\rho\gamma_0}(0)} |u| + \sup_{B^n_{\rho\gamma_0}(0)} |Du| + \sigma^\alpha [Du]_{C^{0,\alpha}(B^n_{\rho\gamma_0}(0))} \leq C \, \delta^{\frac{1}{2n+2}},$$

where

$$[Du]_{C^{0,\alpha}(B^n_{\rho\gamma_0}(0))} = \sup_{\substack{x,y \in B^n_{\rho\gamma_0}(0) \\ x \neq y}} \frac{|Du(x) - Du(y)|}{|x - y|^\alpha}.$$

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